

Devansh Parapalli

+91 8856962057 | Hyderabad, TG, India | hey@parapalli.dev | github.com/DevParapalli | linkedin.com/in/devparapalli

Professional Summary

Backend and distributed systems engineer with end-to-end production ownership across event-driven microservices, Kafka-backed ingestion pipelines, and container-orchestrated platforms on Kubernetes and OpenShift. Founding engineer and technical lead on Cisco's Finance agent platform, from requirements through production operations. GCP Professional Cloud Architect certified.

Work Experience

Systems Engineer

Jul 2025 — Present

Tata Consultancy Services Limited

Hyderabad, India

- **Client: Cisco**; Scaled the team to 8 engineers under my technical lead within 6 months; own requirements, architecture, and production operations. Outcomes reviewed directly by client stakeholders.
 - Reduced case resolution time from **1+ day to under 30 seconds** for **750+ monthly tickets** at **95% autonomous closure** by building **AIT CaseIQ** as an event-driven microservice: used **Kafka** topics for reliable ingestion into auto-scaling containerized runners, decoupling load spikes from processing capacity, with an agentic LLM layer handling classification and structured output delivery.
 - Built the agent framework on **Google ADK** with backward-compatible **A2A** protocol extensions and a codegen layer that lets business users stand up agents themselves; runs **6 tracks** and **8 agents**, onboarding a new track in **4–5 hours**.
 - Automated SOX audit review for **1,000+ person-hours/month** of payroll reconciliation by engineering a Python reconciliation pipeline that diffs PAAT payroll records against Oracle General Ledger accruals, then feeds outstanding variances and incoming email justifications into an LLM scoring layer; eliminating the manual auditor review loop end-to-end.
 - Saved **1,300+ person-hours/month** across **4 FinOps workflows** by building async pipelines with idempotent retry logic and structured error handling: an SOP-driven multi-source report generator (Snowflake, MongoDB, OracleDB), a RAG-backed knowledge chatbot with tool-calling, a natural language to multi-target query translator, and an automated data aggregation and Excel delivery pipeline.
- **Internal: GenAI Center of Excellence**; Led technical architecture across **14 RFPs**, **7 won** totalling **\$450M+ TCV**, delivering architecture documents and working prototypes demoed directly to prospective clients, covering cloud migration, high-availability architecture, and AI infrastructure engagements.
 - Eliminated external API dependency for enterprise model serving (**7B-120B parameters**) by architecting an on-premises **vLLM** and **LiteLLM** inference platform with centralized API gateway routing, RBAC, PII sanitization, and audit logging.

Backend Development Intern

Jun 2024 — Aug 2024

PolymathAI

Nagpur, India

- Cut per-video AI processing time **12x (3 min → 15 sec)** by designing a parallelized inference pipeline across horizontally-scaled containers, processing **14,600+ videos** while optimizing per-instance request caps and cold-start overhead for cost efficiency.

Selected Projects

AIKO: Multi-Modal Document Engine (aiko.parapalli.dev)

Jun 2024 — Present

- Sustained **p95 ingest latency of 3s** and **p99 of 10s** across a multi-modal document pipeline handling PDFs, web content, and multimedia by implementing a custom serialization layer with cloud-native and offline-capable deployment targets.
- Reduced API response latency by **25%** by replacing SSE with custom **WebSocket** duplex streaming for real-time token delivery.
- Expanded ingestion surface to all major browsers by building a cross-browser **WebExtension** with client-side content capture and real-time backend synchronization.

ARA: AI-powered Research Assistant (ara.parapalli.dev)

Jan 2024 — May 2024

- Built a full-stack RAG-based research workspace with inline web retrieval and cited AI responses across SvelteKit, FastAPI, Supabase, and containerized LLM runners; deployed across Vercel, Railway, and Modal.

Education

Government College of Engineering, Nagpur

Nagpur, India

B.Tech, Computer Science and Engineering

Dec 2021 — Jun 2025

CGPA: 8.28/10.0

Technical Skills

Languages: Python, JavaScript/TypeScript, C/C++, SQL, Zig

Frameworks & Libraries: FastAPI, Node.js/Bun, React.js/Next.js

Distributed Systems: Kafka, Event-Driven Architecture, Microservices, Horizontal Scaling, API Gateways

Databases: PostgreSQL w/ pgvector, Redis, MongoDB, OracleDB, Spanner, Snowflake

Cloud & Infrastructure: GCP (Certified Architect), Docker, Kubernetes, OpenShift, Terraform

DevOps & Tools: Git, GitHub Actions, Jenkins CI, SonarQube, Linux

Certifications & Achievements

- **GCP Professional Cloud Architect** (Dec 2025) | **GCP Cloud Digital Leader** (Sep 2025) | **GCP Generative AI Leader** (Sep 2025) | [Credly](#)
- **Publication:** "AI-based Knowledge Organizer for diverse data formats" - IJTE, March 2025